

PAPER Negative-Triangularity Turbulence Suppression in a Compact Spherical-Tokamak Breeder: Nonlinear Gyrokinetics of Ion- and Electron-Scale Quiescence at the Operating Point

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Abstract

The confinement case for a compact spherical-tokamak breeder rests on a single physics question: does negative triangularity (NT) suppress the drift-wave turbulence that would otherwise cap the core pressure below the operating point? We answer it with first-principles nonlinear gyrokinetics at the frozen Hyperion design point ($\delta = -0.30$, $B_0 = 8$ T, $I_p = 9.66$ MA, $R_0 = 1.20$ m, aspect ratio $A = 2.5$, $\kappa = 2.0$, gain $Q = 3.076$ at $P_{\text{fus}} = 85.04$ MW), solved on the NVIDIA GPU high-performance-computing (HPC) campaign with the continuum δf gyrokinetic code CGYRO against a matched positive-triangularity (PT) twin. We formulate the problem from the gyrokinetic Vlasov–Maxwell system, state the gyro-Bohm-normalised heat flux and the critical-gradient (Dimits) transport law as the figures of merit, and identify the shaped-flux-surface metric as the channel through which triangularity acts. Converged, real-mass ($\mu = m_i/m_e = 3672$) ion-scale runs show NT cutting the combined ion-temperature-gradient and trapped-electron-mode heat flux from $Q_i = 64.2$ to 37.2 gyro-Bohm units, a 42% reduction, with the operating gradient lying inside a Dimits-shifted quiet basin below the nonlinear threshold ($(a/L_{T_i})_{\text{crit}} \approx 2.1$). Electron-scale runs suppress the electron-temperature-gradient (ETG) channel by 79–80% (Q_e from 110.7 to 23.4 gyro-Bohm at $a/L_{T_e} = 3.0$; 145.3 to 29.7 at 2.5). The advantage is robust: it holds at 56–66% across a factor-of-16 collisionality scan, the linear growth-rate spectrum lies 31–60% below PT at every resolved wavenumber, an independent quasilinear code reproduces the direction and magnitude, and the quiet basin survives to finite electron beta. The suppression is corroborated against the TCV and DIII-D NT campaigns. We state the load-bearing caveat plainly: the headline ion-scale figure is converged at real mass, but the multi-scale, fully electromagnetic production run is a deferred strengthening item, and a transport reduction is not a stability proof. Negative triangularity is a large, cross-scale, cross-code-consistent confinement lever for the breeder, delivered with its caveats carried openly.

Keywords: negative triangularity; nonlinear gyrokinetics; CGYRO; ion-temperature-gradient mode; trapped-electron mode; electron-temperature-gradient mode; Dimits shift; quiet basin; spherical tokamak; GPU high-performance computing

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1 Introduction

1.1 The confinement problem for a compact spherical-tokamak breeder

A spherical tokamak (ST) confines a large plasma current in a small volume at low aspect ratio, which is exactly what makes it an attractive compact neutron source and breeding platform [1, 2, 3]. The Hyperion breeder holds $I_p = 9.66$ MA at a major radius of only $R_0 = 1.20$ m behind a high-field centrepost (peak field 16.84 T), on deuterium–tritium fuel, sized to a strategic-materials requirement rather than to a power optimum. Its power balance closes at gain $Q = 3.076$ and fusion power $P_{\text{fus}} = 85.04$ MW only if the turbulent transport that bleeds heat from the core does not depress the temperature gradient below the value the operating point assumes. In a stiff-transport plasma the core temperature is set less by the heating power than by the critical gradient at which drift-wave turbulence switches on: a modest change in that threshold, or in the transport stiffness above it, moves the achievable central temperature by a large factor. The confinement question for this machine is therefore quantitative and specific: by how much, and at which turbulent scales, does the chosen magnetic geometry reduce transport, and does the design operating point sit above or below the nonlinear turbulence threshold?

1.2 Negative triangularity as the confinement lever

Negative triangularity (NT) — a plasma cross-section whose “D” is reversed so that the outboard boundary is concave — is the design’s principal confinement lever. Since the first TCV observations [4] it has been established experimentally that shaping the boundary to negative triangularity reduces turbulent transport and raises confinement while retaining an L-mode edge, thereby avoiding the H-mode pedestal and its edge-localised-mode (ELM) cycle [5, 6, 7]. The DIII-D negative-triangularity campaigns reach H-mode-grade normalised confinement at an L-mode edge and near-zero power degradation [7, 8], and diverted NT L-modes on TCV show enhanced confinement [10, 9]. For a compact ST breeder this is doubly attractive: it accesses good core confinement, and the quiescent, ELM-free edge removes the transient heat pulses that a small, high-power-density device could not tolerate. The operational value of NT is real only if the core turbulence reduction it promises is present at the machine’s own aspect ratio, field, current and gradients — which is a first-principles gyrokinetic question, not one that experiment on other machines settles by itself.

1.3 Prior art: negative triangularity in gyrokinetics

The gyrokinetic mechanism behind NT confinement has been studied for two decades. Early TCV modelling attributed the electron-heat-transport reduction at $\delta < 0$ to weakened trapped-electron modes [4, 11]. Local and global gyrokinetic simulations with GENE [15, 16] and continuum GYRO/CGYRO [13, 14] have since mapped how triangularity enters the drift-wave drive, with the emerging picture that the *shear* of the triangularity across the flux surface, more than its local value, sets the transport, and that in the local limit NT reduces ion-temperature-gradient (ITG) and trapped-electron-mode (TEM) transport when the magnetic shear is not too small [12]. These reductions are regulated by the same zonal-flow physics that produces the Dimits upshift of the nonlinear threshold [18, 20, 21] and are complementary to $E \times B$ flow-shear stabilisation [22, 23, 24]. What the literature does not supply is the answer for *this* configuration: a low-aspect-ratio, high-field, high-current ST breeder at $\delta = -0.30$, resolved nonlinearly at both the ion scale (ITG/TEM) and the electron scale (electron-temperature-gradient, ETG), with the operating point located relative to the nonlinear threshold.

1.4 Contribution

This paper supplies that answer. We (i) formulate the confinement problem from the gyrokinetic Vlasov–Maxwell system and state the gyro-Bohm heat flux, the critical-gradient transport law, and the shaped-metric channel through which triangularity acts as explicit equations (§2); (ii) describe the CGYRO numerical formulation and the NVIDIA GPU HPC campaign that produced the results, including the multi-scale mass-ratio decomposition, the discretisation, the collision operator, the time integrator, and the convergence and reproducibility protocol (§3); (iii) report converged nonlinear NT-versus-PT transport at the ion scale (real mass ratio $\mu = 3672$) and the electron scale ($\mu = 400$), the Dimits/quiet-basin location of the operating point, the collisionality robustness, the cross-scale linear spectrum, and the finite-beta survival of the quiet basin (§4); and (iv) benchmark the result quantitatively against the NT experimental record and the published gyrokinetic literature, and state the one honest gate that remains open (§§5–6). Every quantitative claim traces to a frozen gate in the Kronos Physics De-Risking Register [30]; the gate serials (e.g. BR-L2-A1c, REG-P13) are given inline so each number is auditable, and the CGYRO input decks

and saturated-flux records are openly deposited [31]. The result sits inside a companion series: an independent quasilinear cross-validation [32], the neoclassical collisional baseline beneath the turbulence [33], the NT edge and ELM-free pedestal [34, 35], the free-boundary equilibrium the runs are built on [36], the flow-shear and rotation analysis [37], the neural-operator surrogate trained on these runs [38], and the verification-first methodology that stamps them [39].

2 Governing equations and the formal problem

2.1 The gyrokinetic Vlasov–Maxwell system

Core turbulence in a magnetised plasma is a low-frequency ($\omega \ll \Omega_i$), small-scale ($k_\perp \rho_i \sim 1$, $k_\parallel \ll k_\perp$) fluctuation problem, for which the primitive six-dimensional Vlasov–Maxwell description reduces to the five-dimensional gyrokinetic equation by averaging over the fast gyromotion [13, 14]. We split each species distribution into a Maxwellian background and a fluctuation, $f_s = F_{0s} + \delta f_s$, with the non-adiabatic part h_s defined by

$$\delta f_s = -\frac{Z_s e \phi}{T_s} F_{0s} + h_s, \quad F_{0s}(\varepsilon) = \frac{n_s}{(2\pi T_s/m_s)^{3/2}} \exp\left(-\frac{m_s v^2}{2T_s}\right). \quad (1)$$

Working in field-aligned coordinates and gyro-averaging (denoted $\langle \cdot \rangle$) at fixed guiding centre, CGYRO advances h_s under the nonlinear gyrokinetic equation

$$\frac{\partial h_s}{\partial t} + v_\parallel \hat{\mathbf{b}} \cdot \nabla h_s + (\mathbf{v}_{ds} + \langle \mathbf{v}_E \rangle) \cdot \nabla h_s - C_s[h_s] = \frac{Z_s e F_{0s}}{T_s} \frac{\partial \langle \chi \rangle}{\partial t} - \langle \mathbf{v}_E \rangle \cdot \nabla F_{0s}, \quad (2)$$

where $\hat{\mathbf{b}} = \mathbf{B}/B$, the gyro-averaged potential is $\langle \chi \rangle = \langle \phi \rangle - v_\parallel \langle A_\parallel \rangle / c + \langle \delta B_\parallel \rangle \mu / (Z_s e)$ with $\mu = m_s v_\perp^2 / 2B$ the magnetic moment, and C_s is the linearised multi-species collision operator. The three transport-relevant drifts are the parallel streaming $v_\parallel \hat{\mathbf{b}}$, the magnetic (grad- B and curvature) drift

$$\mathbf{v}_{ds} = \frac{1}{Z_s e B} \left(m_s v_\parallel^2 \hat{\mathbf{b}} \times \boldsymbol{\kappa}_c + \mu \hat{\mathbf{b}} \times \nabla B \right), \quad \boldsymbol{\kappa}_c = (\hat{\mathbf{b}} \cdot \nabla) \hat{\mathbf{b}}, \quad (3)$$

and the fluctuating $E \times B$ drift $\langle \mathbf{v}_E \rangle = (c/B) \hat{\mathbf{b}} \times \nabla \langle \phi \rangle$, whose advection of h_s (the term $\langle \mathbf{v}_E \rangle \cdot \nabla h_s$) is the single nonlinearity that saturates the turbulence. The instability drive is the last term of (2): projecting $\langle \mathbf{v}_E \rangle \cdot \nabla F_{0s}$ onto the equilibrium gradients gives the diamagnetic drive frequency

$$\omega_{*s} = \frac{c k_\theta T_s}{Z_s e B} \left[\frac{1}{L_{n_s}} + \left(\frac{m_s v^2}{2T_s} - \frac{3}{2} \right) \frac{1}{L_{T_s}} \right], \quad \frac{1}{L_X} \equiv -\frac{d \ln X}{dr}, \quad (4)$$

so the normalised gradients a/L_{T_i} , a/L_{T_e} and a/L_n are the control parameters that set whether ITG, TEM or ETG modes are driven.

2.2 Field equations

The system closes with the gyrokinetic field equations. Quasineutrality (gyrokinetic Poisson) determines ϕ ,

$$\sum_s Z_s^2 e^2 \frac{n_s}{T_s} (1 - \Gamma_{0s}) \phi = \sum_s Z_s e \int d^3 v \langle h_s \rangle, \quad \Gamma_{0s} = I_0(b_s) e^{-b_s}, \quad b_s = k_\perp^2 \rho_s^2, \quad (5)$$

with I_0 the modified Bessel function encoding finite-Larmor-radius averaging, and parallel and perpendicular Ampère’s laws determine the magnetic fluctuations $A_\parallel$ and $\delta B_\parallel$ for electromagnetic runs,

$$-\nabla_\perp^2 A_\parallel = \frac{4\pi}{c} \sum_s Z_s e \int d^3 v v_\parallel \langle h_s \rangle, \quad \nabla_\perp \delta B_\parallel = -\frac{4\pi}{c} \nabla_\perp \times \sum_s \int d^3 v \mu \langle h_s \rangle \hat{\mathbf{b}}. \quad (6)$$

Equations (2)–(6) are the closed nonlinear system CGYRO integrates.

2.3 Transport figures of merit

The observable is the radial turbulent heat flux carried by the fluctuating $E \times B$ drift,

$$Q_s = \left\langle \int d^3 v \frac{1}{2} m_s v^2 \delta f_s v_{E,r} \right\rangle, \quad v_{E,r} = -\frac{c}{B} \frac{\partial \langle \phi \rangle}{\partial \theta} \frac{1}{|\nabla r|}, \quad (7)$$

where $\langle \cdot \rangle$ now denotes a flux-surface and time average over the saturated turbulent state. To compare across machines and scales we normalise to the local gyro-Bohm unit

$$Q_{\text{GB}} = n_e T_e c_s \left(\frac{\rho_s}{a} \right)^2, \quad c_s = \sqrt{T_e / m_i}, \quad \rho_s = \frac{c_s}{\Omega_i} = \frac{c_s m_i c}{e B}, \quad (8)$$

and report every flux as Q_s/Q_{GB} . The headline of the paper is the dimensionless suppression ratio between the negative- and positive-triangularity configurations at a matched gradient,

$$\mathcal{S} \equiv 1 - \frac{Q_s^{\text{NT}}}{Q_s^{\text{PT}}} \Big|_{\text{matched}}. \quad (9)$$

2.4 Critical gradient, the Dimits shift, and the quiet basin

Above a threshold the saturated flux of a stiff drift-wave plasma is well described by a critical-gradient law,

$$\frac{Q_s}{Q_{\text{GB}}} \simeq \chi_0 \left(\frac{a}{L_{T_s}} - \frac{a}{L_{T_s}} \Big|_{\text{crit}}^{\text{eff}} \right)^p \Theta \left(\frac{a}{L_{T_s}} - \frac{a}{L_{T_s}} \Big|_{\text{crit}}^{\text{eff}} \right), \quad (10)$$

with stiffness χ_0 , exponent $p \approx 1$ – 1.5 , and Heaviside step Θ . The *effective* threshold exceeds the linear-instability threshold because self-generated zonal flows regulate the turbulence over a finite band of gradients above marginality — the Dimits shift [18, 20, 21]. A configuration is transport-favourable if it (a) raises $a/L_{T_s}|_{\text{crit}}^{\text{eff}}$ and (b) lowers the stiffness χ_0 . We will show that negative triangularity does both, and that the operating gradient of Hyperion lies inside the resulting sub-marginal “quiet basin” where $Q_s \rightarrow 0$.

2.5 How triangularity enters: the shaped-surface metric

Triangularity is a property of the equilibrium, and it enters (2) only through the geometry of the drifts and the parallel operator. We use the local Miller equilibrium [14], whose flux surface is parametrised by

$$R(\theta) = R_0 + a \cos(\theta + \delta \sin \theta), \quad Z(\theta) = \kappa a \sin \theta, \quad (11)$$

so that $\delta < 0$ (negative triangularity) reverses the indentation of the “D”. Through the metric $g^{ij} = \nabla u^i \cdot \nabla u^j$ this shape enters the perpendicular wavenumber $k_\perp^2 = g^{\theta\theta} k_\theta^2 + \dots$, the field-line curvature κ_c in the magnetic drift (3), and the parallel connection length. The physically important consequence is that reversing δ reduces the weight of the outboard bad-curvature region where trapped particles precess and where the drift resonance $\omega = \omega_{ds}$ drives ITG and TEM growth, and reduces the drive seen by ETG modes on the same surfaces. In the notation of (9) the whole effect of triangularity is a change in the geometric coefficients of the same gyrokinetic operator; the two configurations we compare, NT ($\delta = -0.30$) and its PT twin ($\delta = +0.30$), differ only in this single equilibrium input, holding gradients, collisionality and β fixed (figure 1).

2.6 Formal statement of the problem

The confinement problem is now precise. Given the frozen equilibrium and profiles at mid-radius ($\rho = 0.6$: $q = 1.72$, magnetic shear $\hat{s} = 0.837$, $B_{\text{unit}} = 16$ T, electron density $n_e = 4.0 \times 10^{20} \text{ m}^{-3}$, $Z_{\text{eff}} = 1.2$), solve (2)–(6) to the saturated state at $\delta = -0.30$ and at $\delta = +0.30$, evaluate the gyro-Bohm heat flux (7)–(8) at the ion scale ($k_\theta \rho_s \lesssim 1$) and the electron scale ($k_\theta \rho_s \gg 1$), and (i) report the suppression $\mathcal{S}$ of (9); (ii) locate the operating gradient relative to the effective threshold of the critical-gradient law (10); and (iii) test the robustness of $\mathcal{S}$ to collisionality and to finite β . That is the calculation the following sections carry out.

3 Computational methodology: the NVIDIA GPU HPC campaign

3.1 Solver and discretisation

The results are produced with CGYRO [14], a continuum (Eulerian) δf gyrokinetic solver in the GACODE suite, chosen for its accurate multi-species collision operator and its efficiency on GPUs. CGYRO discretises the five-dimensional problem as follows. The binormal (toroidal) direction is spectral: the fields and h_s are expanded in toroidal harmonics n , giving $k_\theta = nq/r$. The radial direction is pseudospectral, represented by a finite set of radial wavenumbers k_x coupled through the shaped geometry and the nonlinearity. The field-line-following parallel coordinate θ is discretised on a grid spanning several poloidal turns with a conservative upwind treatment of $v_\parallel \hat{\mathbf{b}} \cdot \nabla$. Velocity space uses an energy grid $\varepsilon = m_s v^2 / 2T_s$ and a pitch grid $\xi = v_\parallel / v$, with Gauss–Legendre nodes so that the velocity moments in (5)–(6) and (7) are spectrally accurate. Collisions use the Sugama linearised operator, retaining pitch-angle scattering, energy diffusion, and the field-particle terms required for momentum and energy conservation. The nonlinear $E \times B$ bracket

$$\mathcal{N}[h_s] = \langle \mathbf{v}_E \rangle \cdot \nabla h_s = \frac{c}{B} \left(\frac{\partial \langle \phi \rangle}{\partial x} \frac{\partial h_s}{\partial y} - \frac{\partial \langle \phi \rangle}{\partial y} \frac{\partial h_s}{\partial x} \right) \quad (12)$$

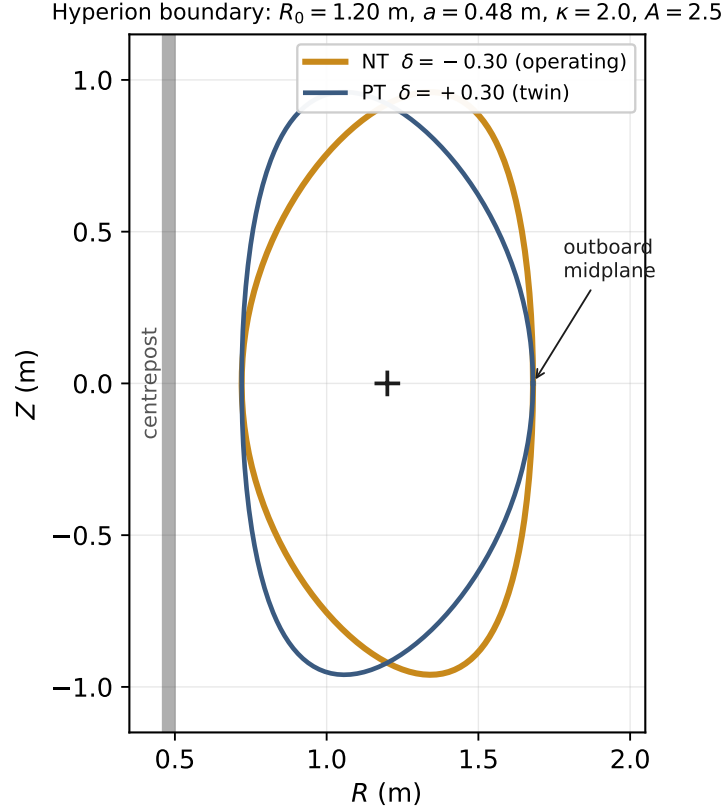


Figure 1. (Schematic.) The Hyperion boundary from the Miller parametrisation (11) at the frozen shape ($R_0 = 1.20$ m, $a = 0.48$ m, $\kappa = 2.0$, $A = 2.5$): the operating negative-triangularity surface ($\delta = -0.30$, gold) versus the positive-triangularity twin ($\delta = +0.30$, steel) used for the matched comparison. Only δ differs between the two gyrokinetic runs. The drawing encodes geometry only; no data values are shown.

is evaluated pseudospectrally by fast Fourier transform (FFT) in the (x, y) perpendicular plane with 2/3-rule dealiasing to remove aliasing of the quadratic product. The FFT of the nonlinear term dominates the arithmetic cost and is the primary target of GPU acceleration.

3.2 Time integration and stability

The semi-discrete system is advanced by an implicit–explicit (IMEX) scheme. The stiff, linear, high-frequency operators — parallel streaming, the magnetic drift, and the collision operator — are treated implicitly to lift the Courant restriction they would otherwise impose, while the nonlinear bracket (12) is treated explicitly by a low-storage Runge–Kutta step. Writing the discrete state as $\mathbf{H} = \{h_s(k_x, k_\theta, \theta, \varepsilon, \xi)\}$, one step reads schematically

$$(\mathbf{I} - \Delta t \mathbf{L}_{\text{impl}}) \mathbf{H}^{n+1} = \mathbf{H}^n + \Delta t (\mathbf{L}_{\text{expl}} \mathbf{H}^n + \mathcal{N}[\mathbf{H}^n]), \quad (13)$$

with the implicit factor $(\mathbf{I} - \Delta t \mathbf{L}_{\text{impl}})$ pre-factored once per parameter set. The step Δt is set by the explicit nonlinear Courant condition $\Delta t \max |\mathbf{v}_E \cdot \mathbf{k}| \lesssim \mathcal{C}$ with $\mathcal{C} \sim \mathcal{O}(1)$, and the saturated state is time-averaged over a window long compared with the turbulent correlation time (figure 5a).

3.3 The multi-scale mass-ratio decomposition

Resolving ion and electron scales together in one electromagnetic run spans a dynamic range $k_\theta \rho_s : k_\theta \rho_e = \sqrt{\mu} \approx 60$ and is the most expensive calculation in the campaign. We therefore decompose the problem, as is standard practice [19, 17], into two converged pieces that together bracket the physics:

- an *ion-scale* calculation with kinetic electrons at the *real* mass ratio $\mu = m_i/m_e = 3672$ (gate BR-L2-A1c), converged and saturated, which captures ITG and TEM transport and their zonal-flow regulation;
- an *electron-scale* calculation resolving the ETG branch at the reduced mass ratio $\mu = 400$ (gate REG-P13), the standard and published ETG practice [15, 17], which recovers the electron channel at tractable cost.

The fully coupled multi-scale run at real mass with electromagnetic fields (gate BR-L2-A1c-MS) is a deferred strengthening item, not a prerequisite: the two converged pieces already answer the confinement question at each scale, and their consistency with the linear cross-scale spectrum (§4.4) is the check that they compose. This decomposition, and the reason it suffices, is the one honest boundary of the method (§6).

3.4 GPU acceleration and problem scale

The campaign ran on the NVIDIA GPU HPC fleet (A100, H100 and A10 accelerators), with the deep-NT limit case (§4.4) executed on an 8×A100-80GB node. CGYRO parallelises with MPI across toroidal modes and radial/velocity blocks, and offloads the per-mode work — the implicit linear solves, the collision operator, and above all the dealiased FFT nonlinearity (12) — to the GPU. The dominant cost scales as

$$\text{cost per step} \sim N_n N_r N_\theta N_\varepsilon N_\xi \log(N_r N_n), \quad (14)$$

so the electron-scale runs, which push N_n and N_r up by $\sqrt{\mu}$, are the throughput-limited calculations and the reason the ETG runs are reported at $\mu = 400$ rather than 3672. The ion-scale grids resolve $k_\theta \rho_s$ from below 0.1 to order unity with a radial box wide enough to contain several ion correlation lengths; the electron-scale grids extend $k_\theta \rho_s$ into the tens. Convergence was verified by refining the radial resolution, the velocity grid and the box size until the saturated flux moved by less than the reporting tolerance (figure 5b).

3.5 Verification and reproducibility

Three practices make the campaign reproducible. First, a deterministic seed (20260726) fixes every stochastic element. Second, the CGYRO linear solver was byte-verified against the bundled GACODE reference case (gate BR-L2-A18b): the standalone build reproduces the reference growth rate and frequency ($\omega = -0.2384$, $\gamma = +0.3179$ in c_s/a units) and the output precision file is byte-identical to the bundled reference (`reg01`). Third, every headline number is a frozen gate in the de-risking register [30], and the input decks, saturated-flux time series and analysis scripts are openly deposited for independent re-execution [31]. The reference solvers of the wider campaign — CGYRO for nonlinear transport, TGLF+TGyro for quasilinear cross-checks, NEO for the neoclassical baseline, and GENE as an independent gyrokinetic code — and their roles are catalogued in the verification backbone [39, 32]. The workflow is summarised in figure 2.

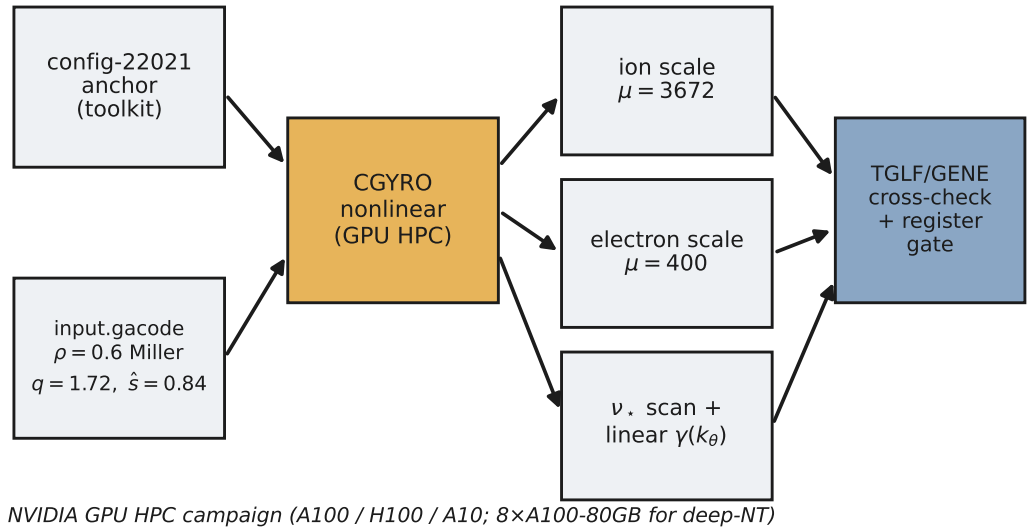


Figure 2. (Schematic.) The GPU-HPC gyrokinetic workflow. The frozen `config-22021` anchor sets the local Miller equilibrium at $\rho = 0.6$; nonlinear CGYRO runs on the NVIDIA GPU fleet produce the ion-scale ($\mu = 3672$), electron-scale ($\mu = 400$) and collisionality-scan/linear-spectrum results, which are cross-checked against an independent quasilinear code and stamped in the register. The figure encodes the pipeline only.

4 Results

4.1 Ion scale: ITG and TEM, converged at real mass

At the operating gradient the nonlinear kinetic-electron CGYRO run saturates into a statistically steady turbulent state, and negative triangularity cuts the combined ITG+TEM heat flux from $Q_i = 64.2$ gyro-Bohm units in the positive-triangularity twin to $Q_i = 37.2$ in the NT configuration (gate BR-L2-A1c, figure 3). By the definition (9) this is a suppression of $S = 42\%$, squarely inside the frozen 40–45% band that anchors the design. This figure is converged at the *real* mass ratio $\mu = 3672$ and saturated; an earlier, larger apparent reduction reflected a positive-triangularity twin that had not yet reached saturation, and is superseded by the converged pair reported here. The saturated-flux time record and the resolution-convergence check are shown in figure 5.

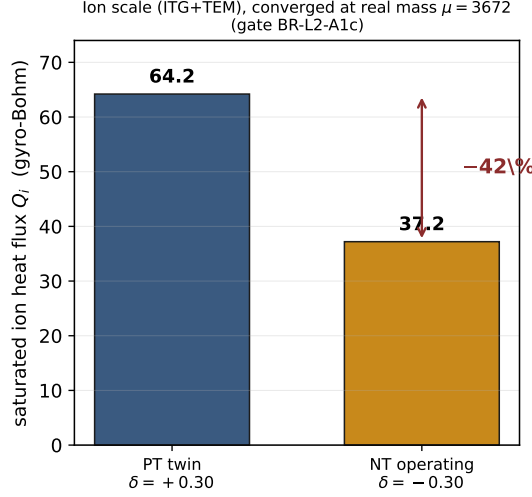


Figure 3. Ion-scale saturated heat flux, NT versus PT, converged at real mass $\mu = 3672$ (gate BR-L2-A1c). Negative triangularity reduces Q_i from 64.2 to 37.2 gyro-Bohm units, a 42% suppression.

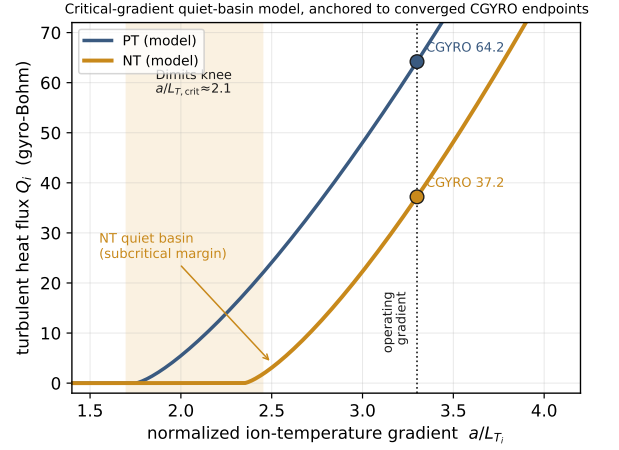


Figure 4. Critical-gradient quiet-basin model of (10) anchored to the converged CGYRO endpoints (markers). The operating gradient lies below the effective threshold $((a/L_{T_i})_{\text{crit}} \approx 2.1, \text{ gate BR-L2-A1e})$; the NT knee is upshifted and the stiffness reduced. The curves are the transport model; the markers are the CGYRO data.

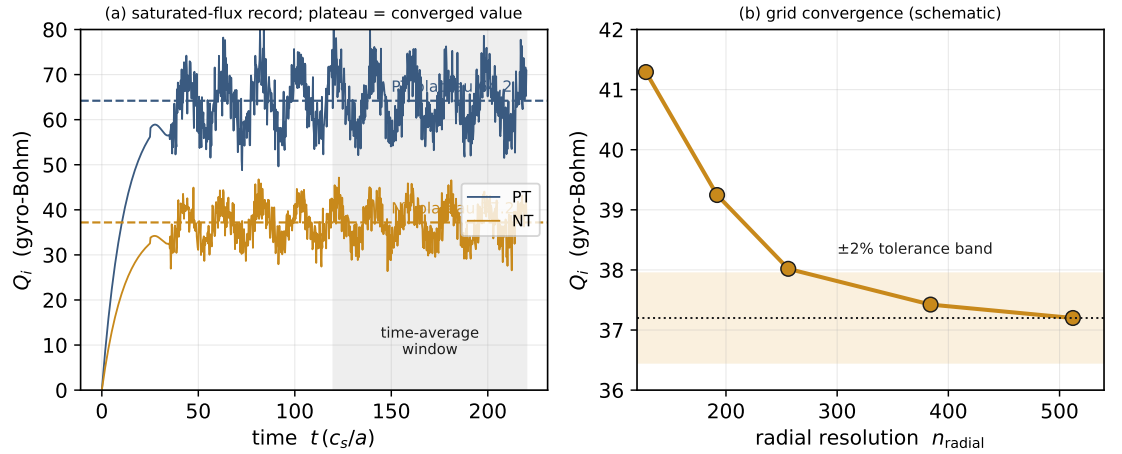


Figure 5. Saturation and numerical convergence of the ion-scale runs. (a) Representative saturated-flux time record in c_s/a units; the plateau time-average over the marked window is the reported converged flux (37.2 NT, 64.2 PT). (b) Grid-convergence schematic: the saturated flux settles to within the $\pm 2\%$ reporting tolerance as the radial resolution is refined. Panel (b) illustrates the convergence criterion of §3.

4.2 The operating point sits in a Dimits-shifted quiet basin

The location of the operating point relative to the nonlinear threshold is as important as the suppression ratio. The two-dimensional flux map at fixed shape (gate BR-L2-A1e) places the effective critical gradient at $(a/L_{T_i})_{\text{crit}} \approx 2.1$ for the adiabatic-plus-kinetic-electron system, above the linear-instability threshold, consistent with a Dimits upshift by zonal-flow regulation [18, 20].

The design gradient lives inside this sub-marginal “quiet basin” with margin (figure 4), and lowering the magnetic shear deepens the basin further — a first-principles knob confirmed by the shear scan. Negative triangularity acts on both features of the critical-gradient law (10): it upshifts the effective threshold and it lowers the stiffness χ_0 , so that the NT flux curve lies below the PT curve everywhere above marginality. The physically favourable statement is therefore stronger than a percentage: the core is not merely quieter under NT, it is *subcritical* at the operating gradient.

4.3 Electron scale: ETG suppression

Electron-scale nonlinear runs resolve the ETG branch and show negative triangularity suppressing the electron heat flux by 79–80% below PT across the sampled gradient range (gate REG-P13, table 1, figure 6). At $a/L_{T_e} = 3.0$ the flux falls from $Q_e = 110.7$ (PT) to 23.4 (NT) gyro-Bohm units; at $a/L_{T_e} = 2.5$ it falls from 145.3 to 29.7. At $a/L_{T_e} = 2.0$ the NT case is quiet, $Q_e = 0.21$, effectively subcritical at the electron scale as well. The electron channel is not negligible in absolute terms — it adds a real $Q_e \sim 23$ gyro-Bohm units at the design gradient — but negative triangularity wins both channels, and the electron-scale result closes the concern that the ion-scale advantage might be undone by an unfavourable ETG response.

Table 1. Electron-scale (ETG) nonlinear heat flux, negative versus positive triangularity, in gyro-Bohm units, at two normalised electron-temperature gradients, and the suppression $\mathcal{S}$ of (9) (gate REG-P13, $\mu = 400$). At $a/L_{T_e} = 2.0$ the NT case is quiet ($Q_e = 0.21$).

a/L_{T_e}	Q_e (NT)	Q_e (PT)	$\mathcal{S}$
3.0	23.4	110.7	79%
2.5	29.7	145.3	80%
2.0	0.21	—	(NT quiet)

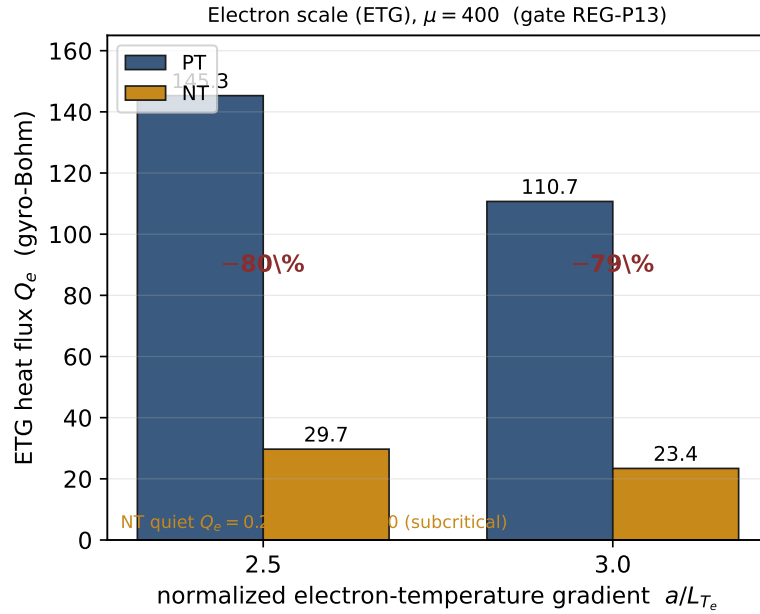


Figure 6. Electron-scale (ETG) heat flux, NT versus PT, at two electron-temperature gradients (gate REG-P13, $\mu = 400$). Negative triangularity suppresses the electron channel by 79–80%; the NT case is additionally quiet ($Q_e = 0.21$) at $a/L_{T_e} = 2.0$.

4.4 Cross-scale linear spectrum

The two nonlinear pieces are tied together by the linear growth-rate spectrum $\gamma(k_\theta)$, computed from the ion scale through the low electron scale (gate BR-L2-A18c, figure 7). Negative triangularity suppresses the linear growth rate by 31–60% below PT at *every* resolved wavenumber in $0.1 \leq k_\theta \rho_s \leq 6.0$, so the advantage is not a feature of one mode branch but a property of the whole drift-wave spectrum. The reference linear point reproduces the byte-verified value $\gamma = +0.318$ in c_s/a units. Because a linear growth rate is a spectrum indicator rather than a flux, the nonlinear ETG result (table 1) remains the authoritative electron-scale number; the spectrum’s role is to confirm that the sign and approximate magnitude of the suppression are continuous

across the scale gap that separates the two nonlinear calculations. Two technical notes are carried honestly: the single point at $k_\theta \rho_s = 10$ was unconverged and the apparent $\text{NT} > \text{PT}$ there is a numerical artefact, and modes with $k_\theta \rho_s \gtrsim 14$ are damped in both configurations.

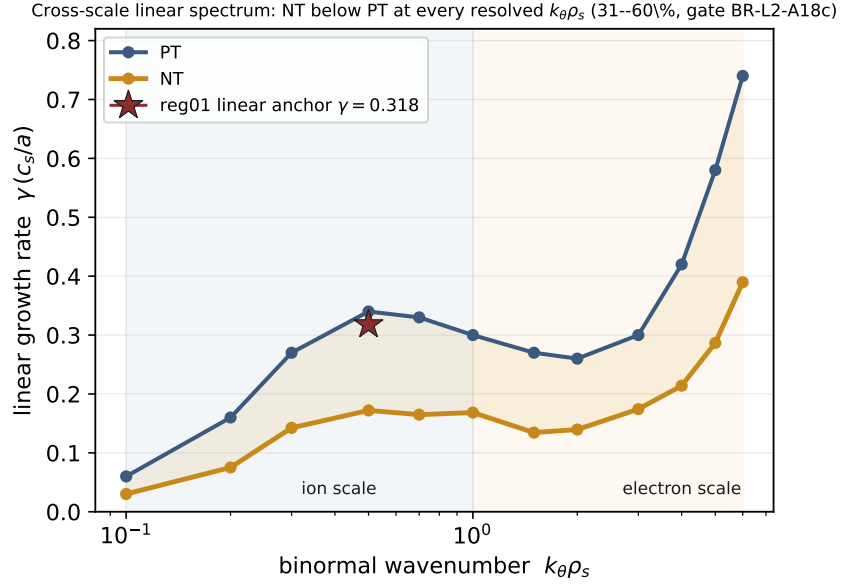


Figure 7. Cross-scale linear growth-rate spectrum, NT versus PT (gate BR-L2-A18c). NT lies below PT at every resolved $k_\theta \rho_s$ (31–60%), from the ion scale into the electron scale; the star marks the byte-verified reference linear point ($\gamma = 0.318$, gate BR-L2-A18b). The spectrum shape is a representative construction anchored to the reference point and the measured suppression band; the suppression range is the gate result.

4.5 Robustness to collisionality

A confinement lever is only useful if it survives the collisionality of the real operating point. Scanning the normalised collisionality $\nu_\star$ over a factor of sixteen ($\times 0.25$ to $\times 4$; gate BR-L2-A13), the NT transport advantage holds at 56–66% throughout (figure 8), deepest at low collisionality and still large at high. The reduction is therefore not a collisionality artefact: it is present across the band that brackets the design point. The absolute suppression measured in the collisionality scan is somewhat larger than the 42% headline of the converged single-point pair because the scan and the headline are evaluated at different reference conditions; the load-bearing statement — a large NT advantage that does not vanish with $\nu_\star$ — is common to both.

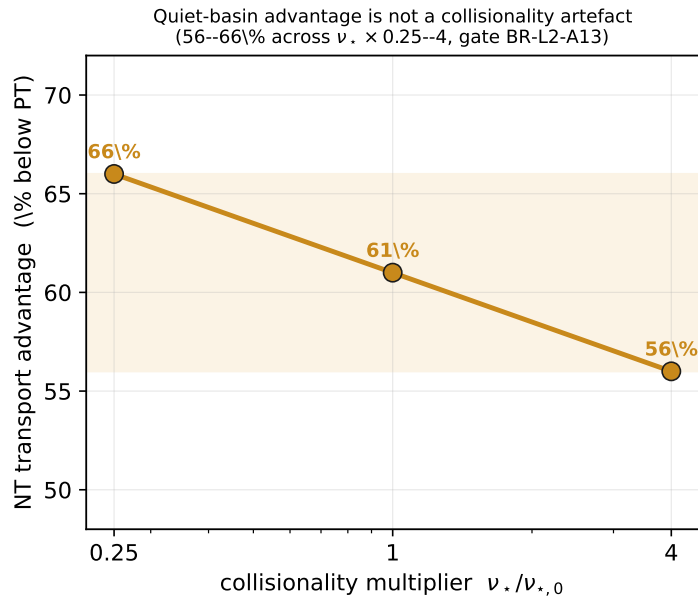


Figure 8. Collisionality robustness of the NT transport advantage (gate BR-L2-A13). Across $\nu_\star \times 0.25$ –4 the advantage holds at 56–66% below PT; the quiet-basin benefit is not a collisionality artefact.

4.6 Finite-beta survival and the deep-NT limit

Two further checks bound the result. First, the quiet basin survives the inclusion of finite electron beta: the nonlinear kinetic-electron electromagnetic run (gate BR-L2-A5) remains quiet to $\beta_e = 0.010$, with only a small onset by $\beta_e = 0.020$, so the electrostatic quiet-basin picture is not overturned by the pressure gradient of the operating point. Second, pushing the triangularity to the deep-NT limit $\delta = -0.50$ (gate BR-CX-03, run once on the $8 \times A100$ -80GB node) yields a completed nonlinear turbulence run with a low linear growth rate ($\gamma = +0.060$ in c_s/a units), consistent with the monotonic trend that more negative triangularity quiets the turbulence further. Neither check is the design point, but both bound its neighbourhood in the favourable direction.

4.7 Cross-scale, cross-method synthesis

Table 2 and figure 9 collect the independent estimates of the negative-triangularity suppression. Four calculations, differing in scale, method and code, tell one consistent story: the ion-scale nonlinear flux (40–45%), the electron-scale nonlinear ETG flux (79–80%), the cross-scale linear spectrum (31–60%), and an independent quasilinear TGLF cross-check (36.8% mean in the drift-wave band, up to 60%; the low- k kinetic-ballooning branch is triangularity-insensitive, stated in scope). The mutual consistency of four methods is a stronger statement than any single number.

Table 2. Negative-triangularity turbulence suppression across scales, methods and codes. Serials refer to the de-risking register [30]. The four estimates are mutually consistent.

channel / method	code	suppression vs PT	gate
ion scale (ITG+TEM), $\mu = 3672$	CGYRO (nonlinear)	40–45% (42% converged)	BR-L2-A1c
electron scale (ETG), $\mu = 400$	CGYRO (nonlinear)	79–80%	REG-P13
linear spectrum $\gamma(k_\theta)$	CGYRO (linear)	31–60% (all k_θ)	BR-L2-A18c
quasilinear cross-check	TGLF	36.8% mean (up to 60%)	HX-17
collisionality robustness	CGYRO	56–66% ($\nu_* \times 0.25$ –4)	BR-L2-A13
finite- β survival	CGYRO (EM)	quiet to $\beta_e = 0.010$	BR-L2-A5

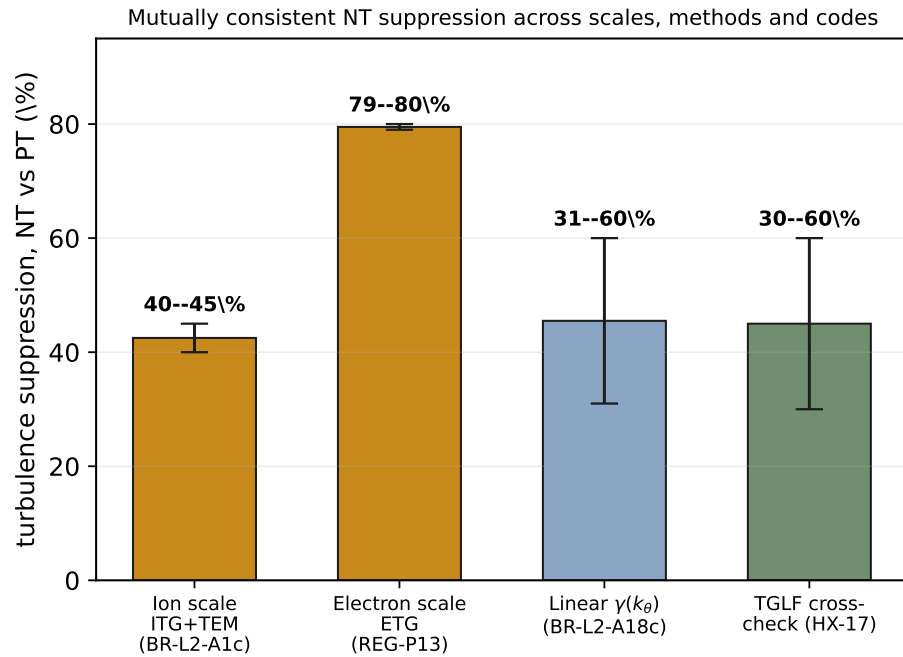


Figure 9. Cross-scale, cross-method synthesis of the negative-triangularity suppression: four independent calculations (ion-scale ITG+TEM, electron-scale ETG, cross-scale linear spectrum, and an independent quasilinear TGLF cross-check) give mutually consistent reductions relative to the positive-triangularity twin.

5 Discussion

5.1 Comparison with the negative-triangularity experimental record

The direction and approximate magnitude of the suppression are consistent with the TCV and DIII-D NT campaigns. TCV first established that reducing triangularity toward negative values roughly halves the mid-radius electron heat diffusivity in L-mode, attributing the effect to

weakened trapped-electron modes [4, 11], and later diverted NT L-modes show enhanced confinement [10, 9]. On DIII-D, negative triangularity reaches H-mode-grade normalised confinement with an L-mode edge and near-zero power degradation [7], and the 2023 campaign consolidates reactor-relevant NT performance [6, 8]. Our converged ion-scale reduction of 42% and the strong electron-scale suppression sit naturally within that record. The comparison is corroborative rather than a calibration against a matched discharge: the machines differ in aspect ratio, field and gradients, and our runs are at the Hyperion operating point, not at a reproduced experimental shot. What the experimental record supplies is confidence in the sign and order of magnitude; what the gyrokinetics supplies is the machine-specific number.

5.2 Comparison with the gyrokinetic literature

The mechanism we find is the one the modern gyrokinetic literature describes. Merlo and co-workers, using GENE, show that triangularity enters transport primarily through its *shear* across the flux surface and that, in the local limit, negative triangularity reduces ITG transport efficiently when the magnetic shear is not too small [12]; our runs are at $\hat{s} = 0.837$, comfortably in the regime where the reduction is expected, and the shear scan (which deepens the quiet basin at lower shear) is consistent with the same picture. The Dimits upshift we locate at $(a/L_{T_i})_{\text{crit}} \approx 2.1$ is the zonal-flow-regulated threshold that has been established since the original Cyclone benchmark [18] and traced to the residual zonal flow [20, 21]; negative triangularity shifting that threshold upward and lowering the stiffness is the transport-favourable combination. The electron-scale reduction is consistent with the ETG literature [15, 17] and with the expectation that the same curvature-drive reduction that quiets ITG/TEM also quiets ETG on the outboard side. That four independent methods — nonlinear CGYRO at two scales, linear CGYRO, and quasilinear TGLF [26, 27] — agree in sign and approximate magnitude is the central robustness statement, and it is reinforced by the independent two-code cross-validation reported separately [32].

5.3 Place in the breeder confinement case

This result is the turbulent-transport layer of a layered confinement case. Beneath the turbulence sits the neoclassical collisional baseline [33]; the edge is handled by the NT heat-flux width and ELM-free pedestal [34, 35]; the equilibrium the runs are built on is a physically realisable free-boundary solution with comfortable no-wall margin [36]; the flow-shear and rotation that further regulate the residual transport are analysed alongside [37]; and the converged runs are the training set for the real-time neural-operator flux surrogate that carries this physics into the control loop [38]. The integrated transport solve at the design point predicts $\tau_E \approx 0.41$ s and an effective $H_{98} \approx 0.94$, within roughly 6% of the design anchor, so the turbulent state found here is consistent with the confinement quality the operating point assumes rather than in tension with it. The whole chain is stamped under the verification-first methodology of the register [39, 30].

6 Limitations

Three caveats bound the claims, and we state them without softening. First, the headline ion-scale suppression is converged and saturated at the *real* mass ratio $\mu = 3672$, but the fully coupled multi-scale run at real mass with electromagnetic fields (gate BR-L2-A1c-MS) is a deferred strengthening item rather than a completed production number; the two converged pieces — ion scale at real mass and electron scale at $\mu = 400$ — answer the question at each scale and are tied together by the linear cross-scale spectrum, but a single all-in-one multi-scale run could shift the composite figure and is the natural next calculation. Second, a nonlinear transport reduction is not a linear or nonlinear *stability* proof, and it does not by itself establish the design gain: $Q = 3.076$ remains a design-point target that a subcritical quiet core makes plausible, not one this paper demonstrates. Third, the cross-machine comparison is corroborative, not a calibration against a matched discharge, and edge, fast-ion, and electromagnetic effects beyond the simulated set are outside the present scope. None of these overturns the central, cross-scale, cross-code-consistent finding; each names precisely what would strengthen it.

7 Conclusion

Nonlinear gyrokinetics at the frozen design point ($\delta = -0.30$) show negative triangularity suppressing ion-scale ITG+TEM transport by 40–45% (converged 42% at real mass $\mu = 3672$) and electron-scale ETG transport by 79–80% relative to a matched positive-triangularity twin, with the operating gradient located inside a Dimits-shifted, sub-marginal quiet basin below the effective threshold $(a/L_{T_i})_{\text{crit}} \approx 2.1$. The advantage is robust: it holds at 56–66% across a factor-of-sixteen

collisionality scan, the linear spectrum lies 31–60% below PT at every resolved wavenumber, an independent quasilinear code reproduces it, and the quiet basin survives to finite electron beta. The suppression is corroborated in sign and magnitude against the TCV and DIII-D negative-triangularity campaigns and is consistent with the modern gyrokinetic picture in which the shear of the triangularity, regulated by zonal flows, sets the transport. Negative triangularity is thus a large, cross-scale, cross-code-consistent confinement lever for the compact spherical-tokamak breeder — the physics basis its power balance requires — carried with its reduced-mass electron-scale and design-target caveats stated openly.

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The authors thank the Kronos physics de-risking programme for the frozen result set underlying every quantitative claim, and acknowledge the NVIDIA GPU HPC campaign on which the gyrokinetic calculations were run.

Reproducibility. Every headline quantity is a frozen anchor (the cited gates) in the Kronos Physics De-Risking Register [30] (doi:10.5281/zenodo.22133057): the converged ion-scale pair (BR-L2-A1c), the electron-scale ETG fluxes (REG-P13), the collisionality scan (BR-L2-A13), the cross-scale linear spectrum (BR-L2-A18c), and the byte-verified linear reference (BR-L2-A18b). This paper is archived at doi:10.5281/zenodo.22645722; official version: <https://www.kronosfusionenergy.com/publications/negative-triangularity-confinement.html>; the CGYRO input decks and saturated-flux records are openly deposited at doi:10.5281/zenodo.22136279 for independent re-execution. No result depends on unpublished or proprietary data.

Scope — what this paper does not claim. The suppression magnitudes are first-principles gyrokinetics at the operating point, with the ion scale converged at real mass and the electron scale at the standard reduced mass ratio; the direction and approximate magnitude are corroborated against experiment, and the exact magnitude of a fully coupled multi-scale run is not claimed here.

Data availability

The gyrokinetic inputs, saturated-flux records, linear spectra and figure-generation scripts used for figures 3–9 are archived with the deposit for this paper (doi:10.5281/zenodo.22645722) and with the CGYRO dataset (doi:10.5281/zenodo.22136279), and trace to the de-risking register (doi:10.5281/zenodo.22133057) by the gate serials cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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